Table 3. Comparison of model performance when using critical tokens versus random tokens.

Method	GSM8K	GSM-Hard	Math500	SVAMP	ARC
Chain-of-Thought	77.40	28.00	31.00	83.00	58.91
random tokens	82.08	40.29	42.12	84.77	75.68
critical tokens (Ours)	**88.24**	**49.73**	**50.00**	**89.60**	**88.31**

Table 4. Comparison of our proposed paraphrase consistency against the majority voting.

Method	GSM8K	GSM-Hard	Math500	SVAMP	ARC
Chain-of-Thought	77.40	28.00	31.00	83.00	58.91
majority voting	87.20	47.36	48.19	88.80	86.16
paraphrase consistency (Ours)	**88.24**	**49.73**	**50.00**	**89.60**	**88.31**

4. Experiments

In this section, we first describe the experimental setup, followed by the main results of our proposed method compared to the baselines. We also perform ablation study and computational cost analysis.

4.1. Setup

Datasets. To comprehensively assess our method, we evaluate it on seven benchmarks. Six focus on mathematical reasoning, including GSM8K (Cobbe et al., 2021), GSM-Hard (Gao et al., 2023), SVAMP (Patel et al., 2021), Math500 (Hendrycks et al., 2021), and the more challenging competition-level datasets AIME2024, AIME2025, BRUMO2025, and HMMT2025 (Balunović et al., 2025). In addition, we use ARC-Challenge (Clark et al., 2018) to evaluate knowledge reasoning ability of LLMs.

Baselines. In our experiments, we use Chain-of-Thought (CoT) (Wei et al., 2022), Self-Consistency (Wang et al., 2023), Tree-of-Thought (ToT) (Yao et al., 2023), Guided Decoding (Xie et al., 2023), Predictive Decoding (Ma et al., 2025), and Phi-Decoding (Xu et al., 2025) as baseline methods.

Metric. Following prior work, we adopt pass@k (k=1,4) as the primary evaluation metric.

Implementation Details. In our experiments, we adopt Llama-3.1-8B-Instruct (Grattafiori et al., 2024), Mistral-7B-Instruct-v0.2 (Jiang et al., 2023), Qwen-3-32B (Yang et al., 2025a) and DeepSeek-R1-Distill-Llama-70B as the target models. we employ the non-thinking mode for Qwen-3-32B. Throughout our method, we employ the same model for generating paraphrased problems, identifying critical tokens, and producing new rollouts. In the first stage, we generate 4 paraphrased variants for each problem in the math benchmarks and 3 variants for each problem in the ARC dataset. In the second stage, we select the top 10 tokens for new rollouts, with the temperature scaling coefficient λ set to 2. For fair comparison, we ensure a comparable inference budget across methods. Specifically, we rollout 48 samples for Self-Consistency (Wang et al., 2023). For Predictive Decoding (Ma et al., 2025) and Phi-Decoding (Xu et al., 2025), we rollout 4-8 samples per foresight step, and each problem typically involves 5–8 foresight steps. We also adopt a zero-shot CoT prompt to elicit the new rollouts. For the baselines, we strictly follow their original settings, including temperature values, sampling strategies, and the number of few-shot examples. All experiments are conducted on NVIDIA A100 GPUs.

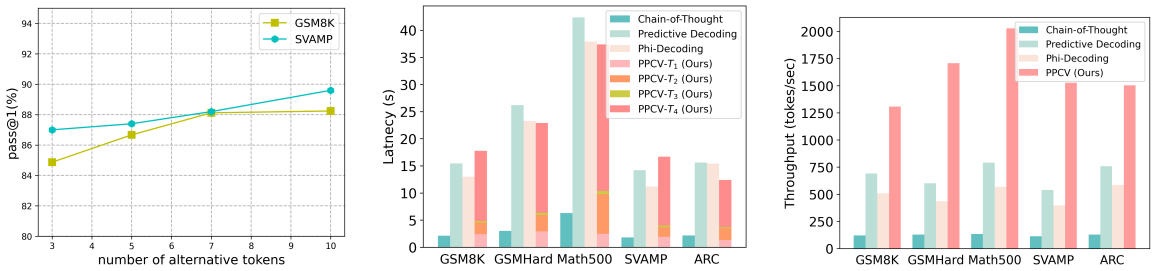
4.2. Main Results

The main results are summarized in Table 1 and Table 2. The results indicate that Self-Consistency effectively improves the reasoning performance of LLMs compared to vanilla Chain-of-Thought prompting. For example, Llama-3.1-8B-Instruct (Grattafiori et al., 2024) achieves about 3% higher accuracy with Self-Consistency than with CoT. These findings suggest that augmenting reasoning during inference through sampling is an effective way to refine reasoning trajectories. Recent decoding-based methods, such as Predictive Decoding (Ma et al., 2025) and Phi-Decoding (Xu et al., 2025), also achieve strong results.Unlike prior works that rely on carefully designed prompts to self-correct errors in intermediate steps, these two methods modify the current step by probing future steps with pre-defined reward signals. Furthermore, our experimental results demonstrate that the proposed method consistently outperforms the baselines across most tasks, spanning both mathematical and knowledge reasoning, thereby highlighting its generalization ability across different reasoning settings. Notably, our method even surpasses the latest approaches such as Predictive Decoding (Ma et al., 2025) and Phi-Decoding (Xu et al., 2025). In particular, it achieves approximately 50.00% accuracy on the Math500 dataset (Hendrycks et al., 2021), exceeding these baselines considerably. The results on competition-level datasets further demonstrate the effectiveness of our method in enhancing the reasoning ability of LLMs. These results indicate that our method can effectively extract critical tokens that play a pivotal role in the final outcome and correct the reasoning trajectory by leveraging alternative tokens. Additional results can be found in Appendix E.

4.3. Ablation Study

In this section, we analyze the contribution of each stage individually. All the evaluations are conducted on Llama-3.1-8B-Instruct (Grattafiori et al., 2024).

Effectiveness of extracted critical tokens. To demonstrate the effectiveness of our extracted critical tokens, we conduct an evaluation in which the critical tokens are replaced with random tokens in the first stage, while keeping the second stage unchanged. This evaluation is performed across multiple benchmark datasets, with pass@1 as the metric. The results, shown in Table 3, reveal a substantial decline in



(a) *(b)* *(c)*

Figure 7. (a) The impact of number of sampled alternative tokens on the performance. (b) Latency comparison between the baselines and our method, measured as the average inference time per question (in seconds). T_1, T_2, T_3, T_4 denote time for paraphrased question generation, initial answer generation, forward pass and new rollouts from truncated trajectories. (c) Throughput comparison between the baselines and our method, measured in output tokens per second.

performance. These findings highlight the pivotal role of critical tokens and indicate that our method can effectively identify and extract them. More ablation study on comparison with Paraphrased Majority Voting (PMV) can be found in Appendix F.

Effectiveness of paraphrase consistency. We also evaluate the effectiveness of our proposed paraphrase consistency and compare it with traditional majority voting. While keeping the first stage unchanged, instead of using paraphrased forms to generate new reasoning steps, we simply sample multiple new steps from alternative tokens conditioned on the original question and use majority voting to determine the final answer. The results, shown in Table 4, reveal a noticeable decline in performance, highlighting the importance of paraphrased forms in improving the intermediate reasoning steps.

Impact of number of sampled alternative tokens.We investigate the influence of the number of sampled alternative tokens in the second stage by selecting values of 3, 5, 7, and 10. The results, shown in Figure 7a, demonstrate that performance improves as the number of alternative tokens increases. This suggests that exploring more reasoning steps with additional alternative tokens during inference can be beneficial for reasoning tasks.

5. Computational Cost Analysis

In this section we examine the composition of the latency in our method. The latency arises from four components: Paraphrased question generation (T_1); initial answer generation (T_2), equivalent to vanilla CoT; a forward pass for identifying critical tokens (T_3), which does not generate new tokens and is computationally lightweight; rollouts of truncated trajectories using alternative tokens under both the original and paraphrased questions (T_4), which constitutes the main source of overhead.

We evaluate all components on Llama-3.1-8B-Instruct using vLLM on NVIDIA A100 GPUs, with a maximum output length of 4096 tokens for each question. For our method, we use 4 paraphrased questions on math datasets and 3 on ARC, and select the top-10 candidate tokens as alternatives. The updated average latency results are reported in Figure 7b. As expected, T_1 scales with the number of paraphrases, T_3 remains minimal, and T_4 dominates the total cost. Specifically, T_4 depends on the number of top-k alternative tokens, the number of paraphrased questions and the position of the critical token in the trajectory. Since the new rollouts from truncated trajectories for different alternative tokens and paraphrased questions are independent, T_4 can take advantage of vLLM's parallelism. These rollouts can therefore be processed concurrently, improving overall efficiency. This is reflected in the higher throughput (tokens/sec) shown in Figure 7c. And results of our method in latency comparable to baseline methods, even on challenging benchmarks such as Math500 and GSM-Hard where the critical token tends to occur in later reasoning steps. On the GSM8K and SVAMP benchmarks, our method, as well as baselines such as Predictive Decoding, would incur a approximately 6–8$\times$ latency overhead compared to vanilla Chain-of-Thought. More analysis on the trade-off between the latency and performance can be found in Appendix G.

6. Conclusion

In this study, inspired by beneficial impact of paraphrase forms on reasoning, we investigate the pivotal role of critical tokens in shaping the reasoning trajectory. To leverage these two factors, we propose the Paraphrastic Probing and Consistency Verification framework. Our framework consists of two stages: Paraphrastic Probing, which identifies and extracts critical tokens, and Consistency Verification, which adopts paraphrase forms to generate new reasoning trajectories with alternative tokens to reach the final answer. We evaluate our proposed framework with different LLMs and extensive evaluations across multiple benchmarks demonstrate the promising performance of our method.

Impact Statement

This paper presents work whose goal is to advance the field of machine learning. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

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